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Deutsche Telekom Chair of Communication Networks

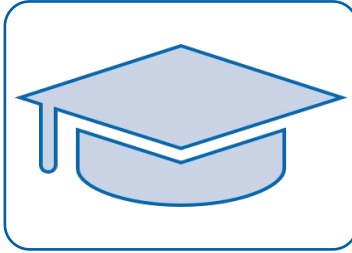
PI: Prof. Dr. Frank Fitzek, Prof. Dr. Ing.- Michael Kaliske, Dr. Stanislav Mudriievskyi

International Tutor: Prof. Dr. Christian Bettstetter

Reliable and Resilient communication for Unmanned Aerial Vehicle in Advanced Air mobility (AAM)

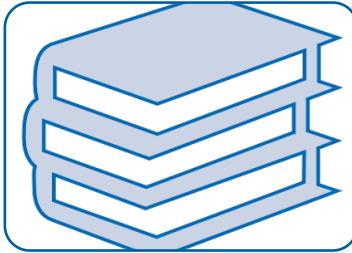


Introduction



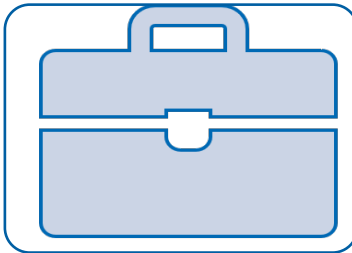
Qualification

- B.Tech Electronics and Communication, SRMIST, India
- M.Sc. Communications Engineering , TU Munich



Relevant Experience

- 2.4 years research assistant at Atmosphere GmbH, Weßling, Munich (Evaluation on Iridium Satellite Communication for UAS)
- Master Thesis: MAVLink Protocol enhancements for UAV Command and Control communication in integrated multi-link scenario (GLOBECOM 2024)



Current position

- PhD in RTG 2947 AirMetro
- Research area : **T9 – Wireless Communication Networks @ Deutsche Telekom Chair of Communication Networks**

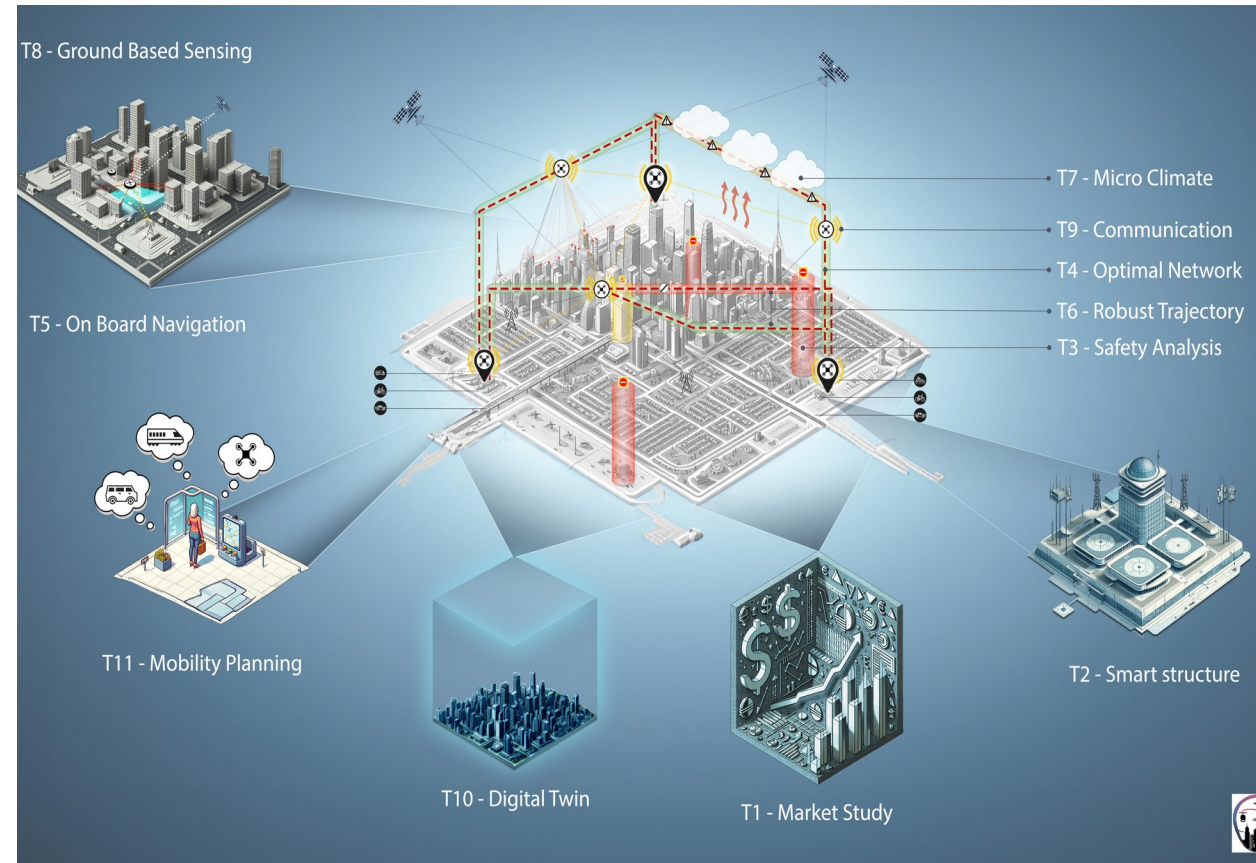
AirMetro webpage: <https://tu-dresden.de/bu/verkehr/airmetro/team/phd-candidates>

Comnets Webpage: <https://cn.ifn.et.tu-dresden.de/chair/staff/m-sc-shreeja-sridharan/>

T- Topic

RTG - AirMetro

- 11 PhD topic
 - 1 Post Doc topic on AAM Planning
 - 10 Principal Investigators
 - 3 Research Clusters
- A: Infrastructure, Cost and Planning
- B: Safety, Micro-Climate and Path Planning
- C: Technological Framework



Technological Framework: CNS

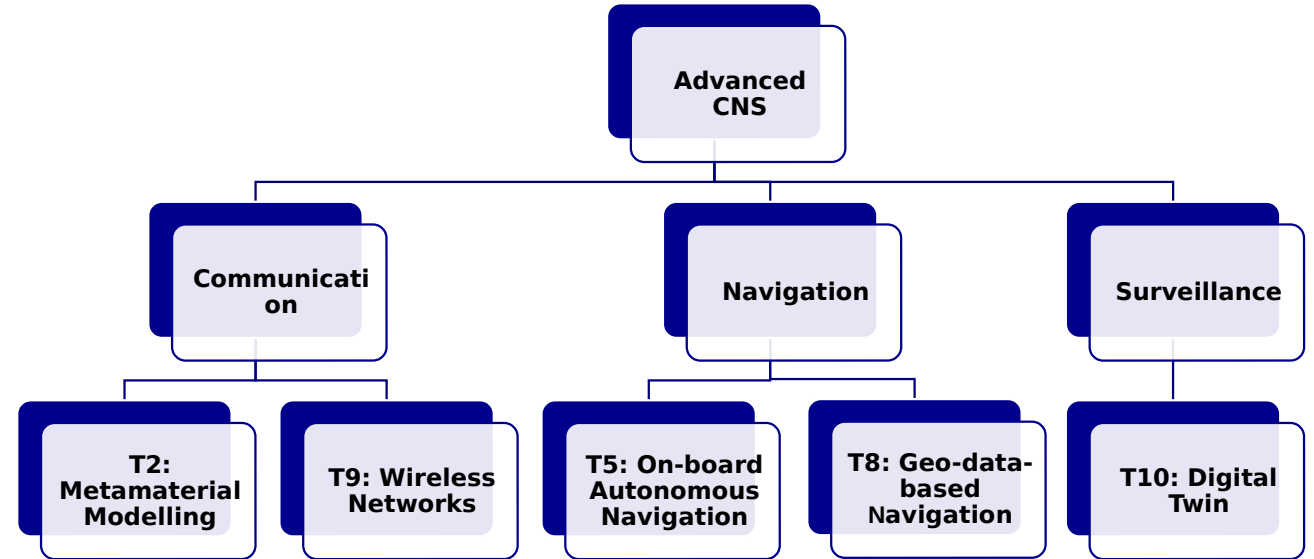
CNS: Communication, Navigation and Surveillance

Three major challenges for AAM technology

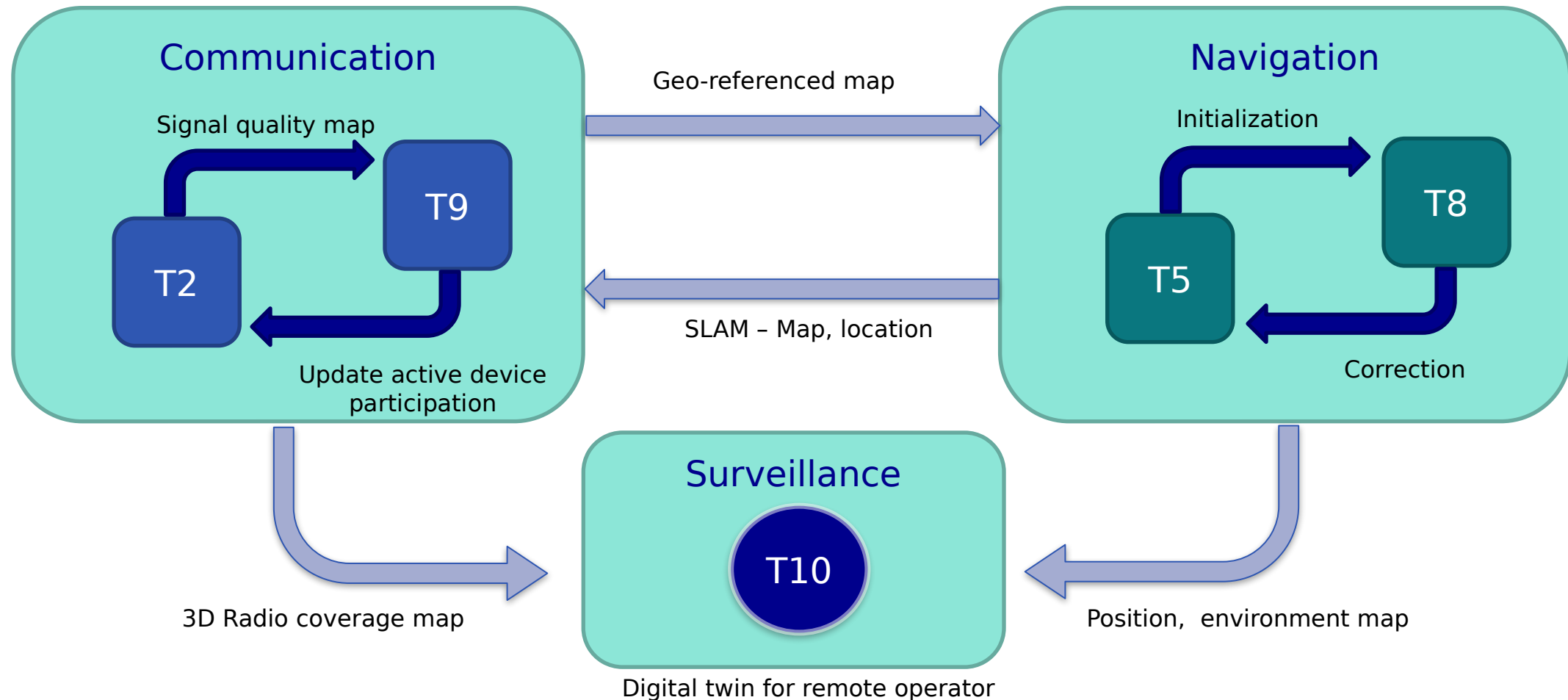
- Very Low-Level airspace (VLL)
- High density of airspace users
- High proximity to infrastructure

Objective

Technological framework enabling safe, reliable autonomous operation



CNS : Research Collaboration



Communication in CNS for AAM

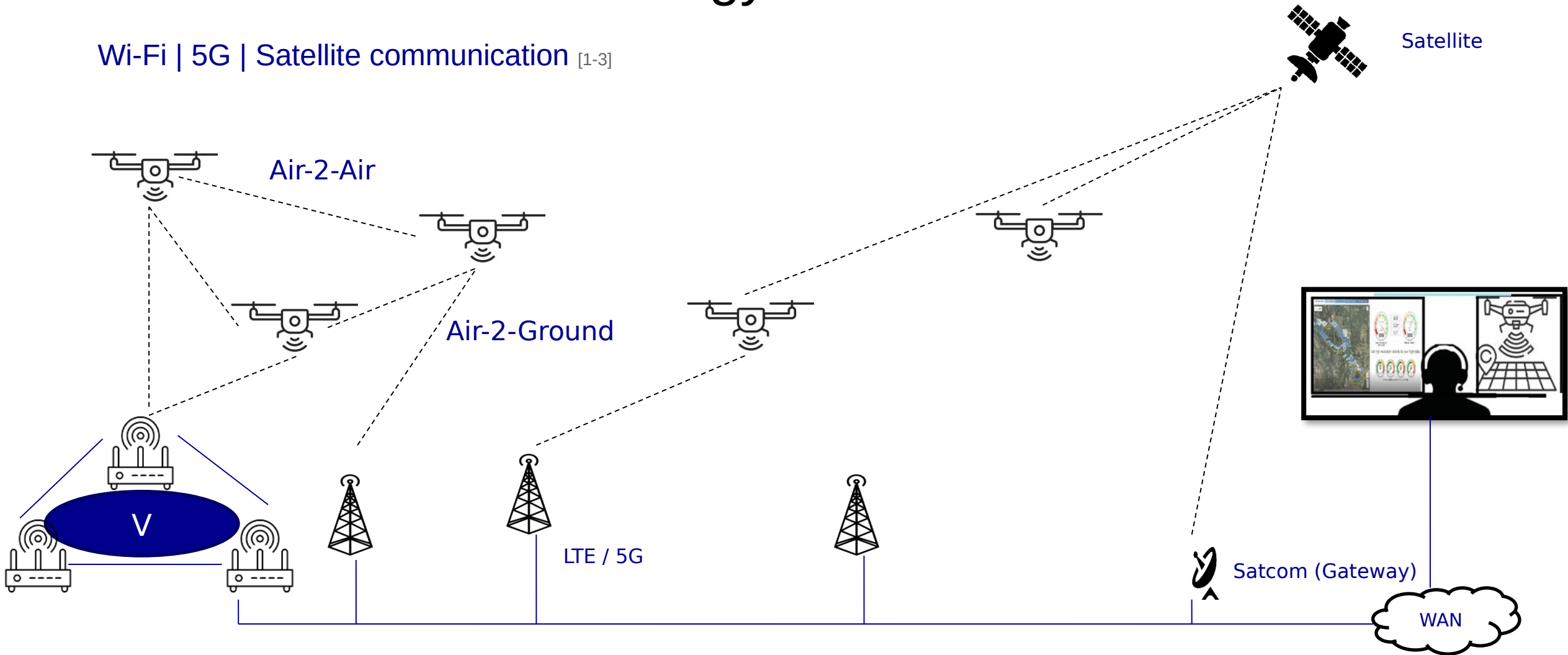
- Communication is the invisible critical backbone for Beyond Visual Line of Sight operation in AAM



- Communication goals encompass ground as well as aerial systems

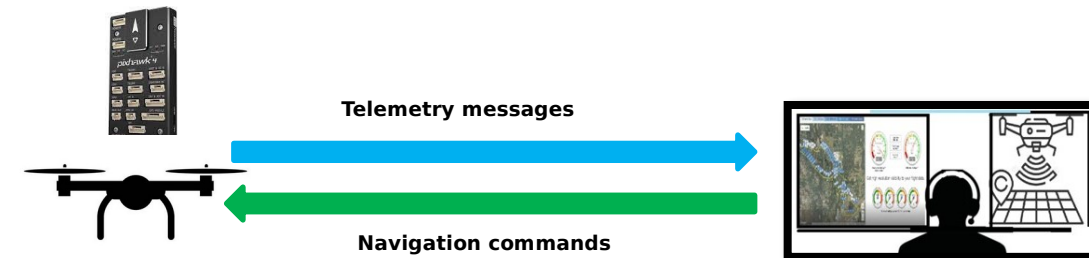
Communication Technology

Wi-Fi | 5G | Satellite communication [1-3]

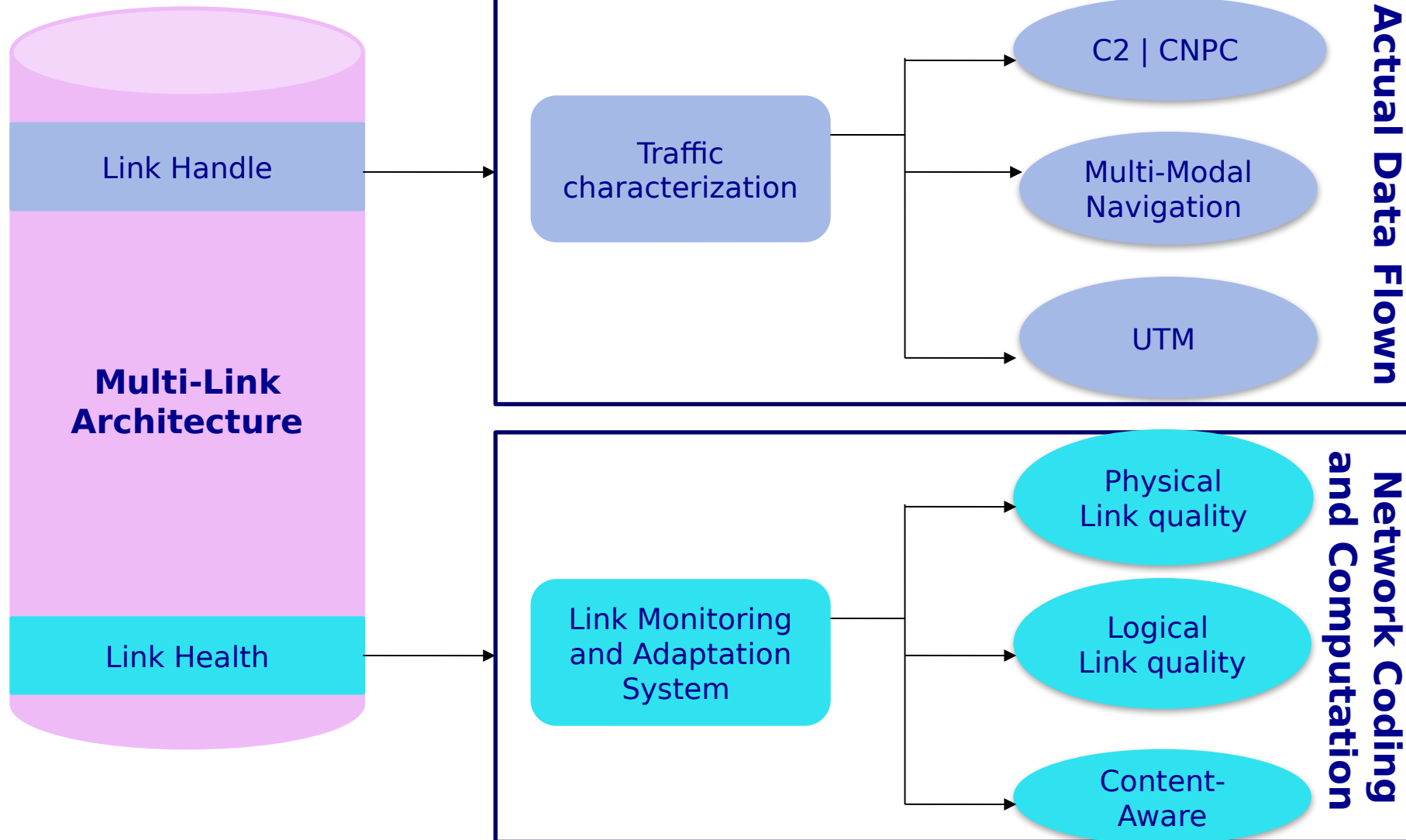


Goal: Reliable and Resilient Communication

- System consist of UAV, GCS, **Command and Control (C2)** and other payload information for safe Beyond Visual Line of Sight (BVLOS) operation [4]
- **Uplink - Telemetry (battery, Lat., Lon., waypoints)**
- **Downlink - Commands (control, speed, altitude)**
- C2 communication also referred as Control and Non-Payload Communication (CNPC)
- Evaluating Air-2-Ground Link
- Addressing Air-2-Air link for emergency messaging



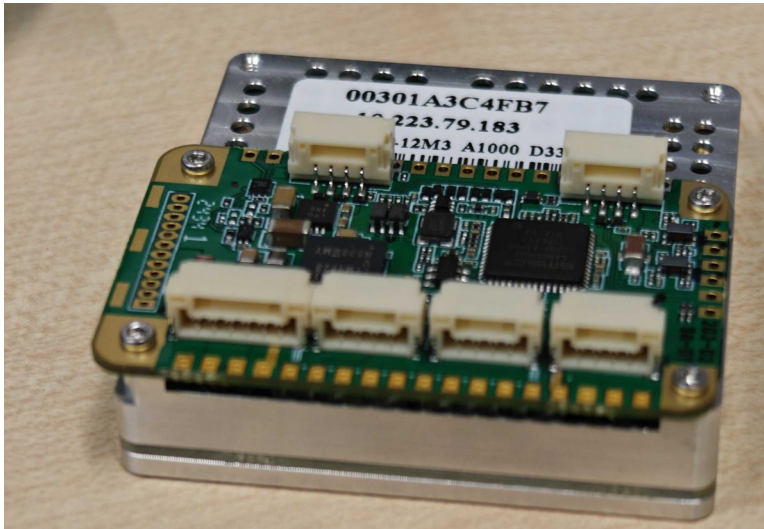
Methodology



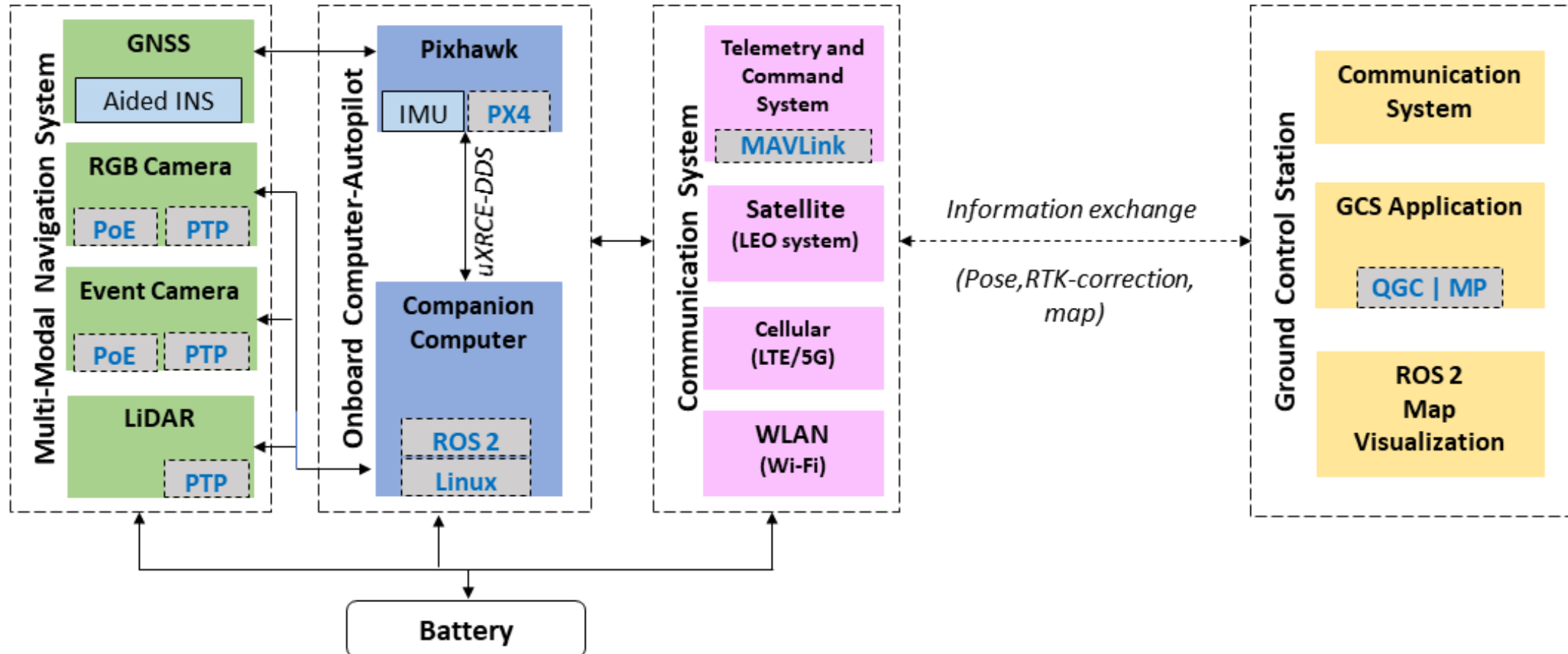
*C2: Command and Control
CNPC: Control and Non-Payload
UTM: Unmanned Traffic
Management*

Plan of Action

- Stage 1: Simulation and Key Metrics, Parameters identification
- Stage 2: Testbed and Flight test [5]
- Stage 3: RLNC based Multi-link protocol



TestBed – Data collection

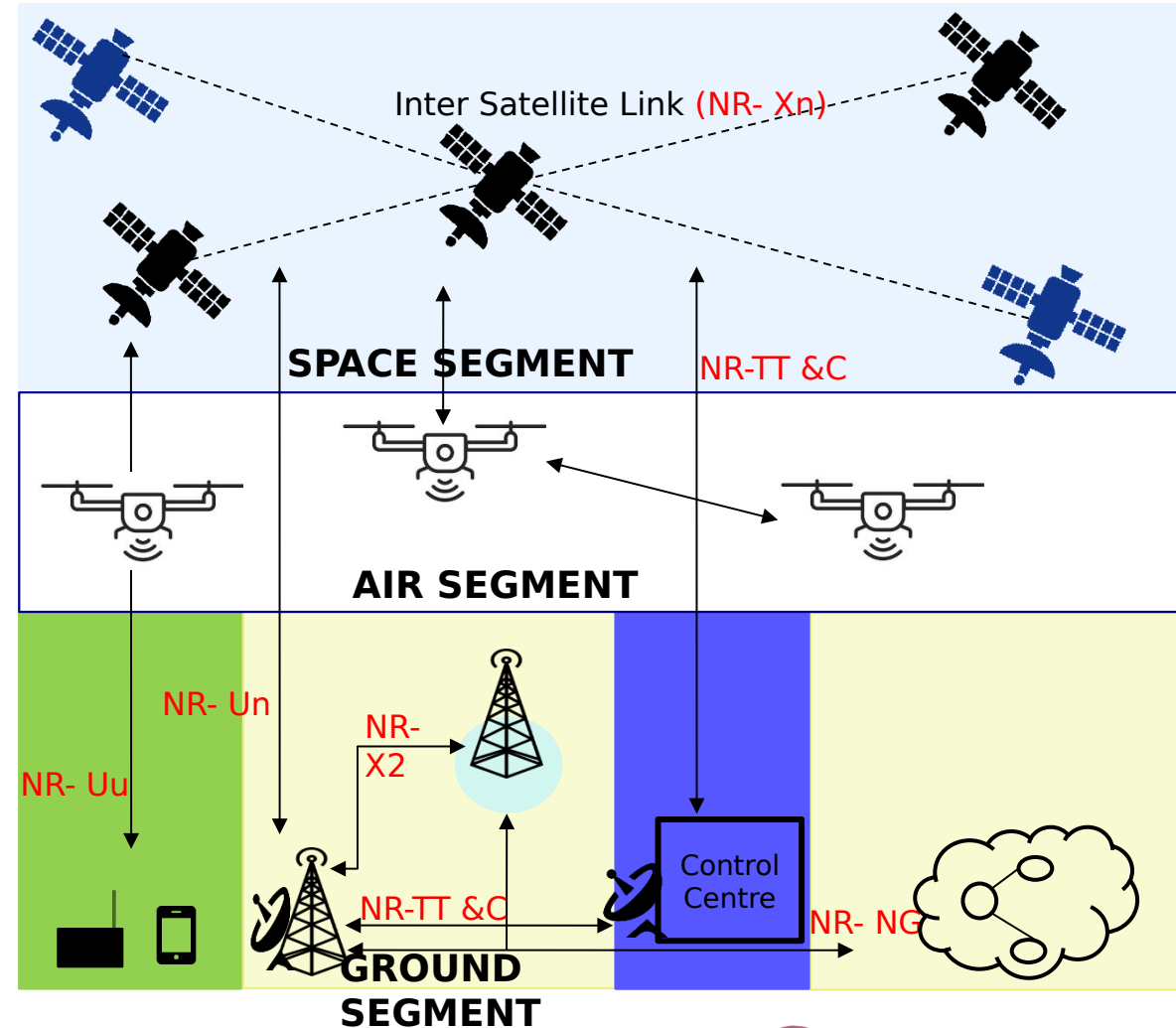


[5]

Research Stay at Uni Klagenfurt

Simulation

- Omnet++ based Link Level Simulation
- Iridium Constellation based LEO Satcom
- Simu 5G NR Network
- Implementation of Users - “Air Segment”
- **Non-telemetry setup**
- Exchange / understanding with system-level simulator @ Mobile Robotics team

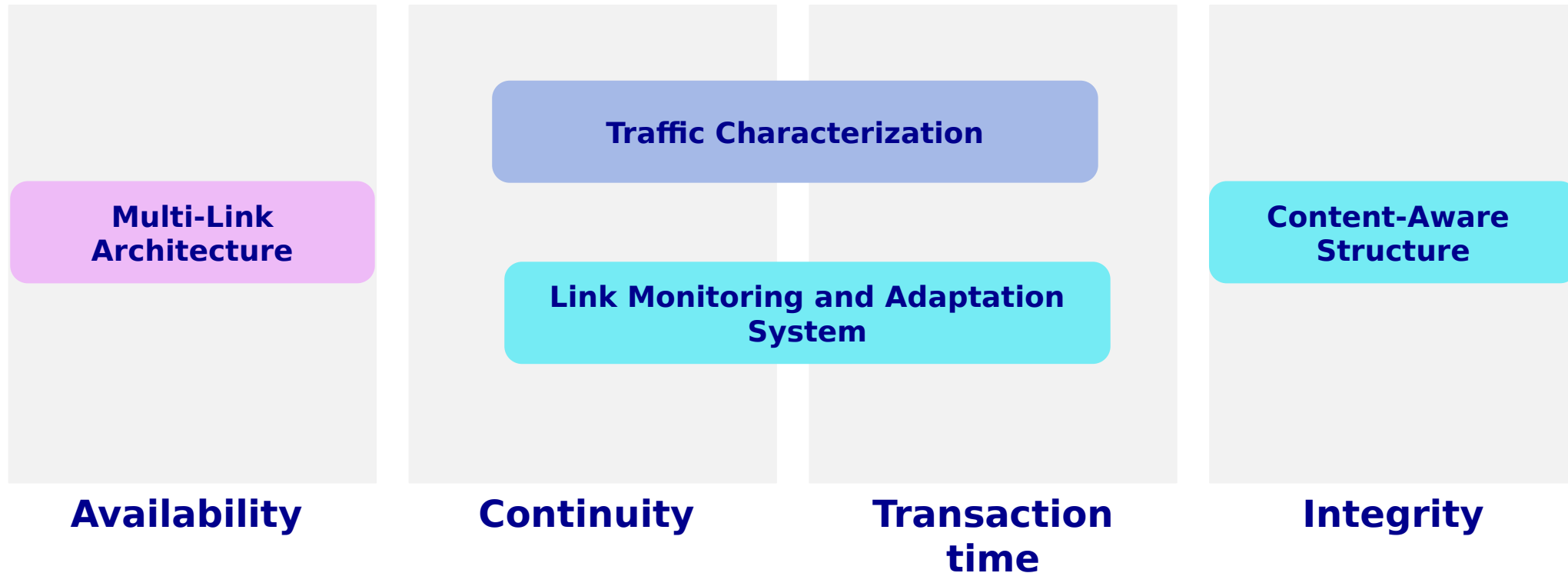


Research Focus - 1/3

- Fine-tune the KPIs and parameters in the simulator
- C2 message characteristics
- Radio based approach – height classification on signal quality
- Implementation and testing
- Work package 1: Implementation of Air user
- Work package 2: Evaluation of the link metrics in Omnet++

Research Focus - 2/3

ICAO Required Communication Performance = Availability, Continuity , Transaction Time, Integrity [6,7]



Research Focus - 3/3

- Intra handover in Cellular link at various altitudes
- Establish requirements of Intra handover (heterogenous multi-link)
 1. Cross-layer optimization
 2. Coordination and Synchronization

Planned Outcomes

- Complete work packages – Results analysis (Bandwidth, Low latency constraints)
- Collaboration publication
- Further exchange on testbed and architecture prototypes

References

- [1] Meshmerize Website news , <https://meshmerize.net/automated-drone-mesh-network/>
- [2] Telekom BMBF project, <https://bmdv.bund.de/SharedDocs/DE/Artikel/DG/mfund-projekte/vision.html>
- [3] Iridium white paper, <https://www.iridium.com/monitored-bvlos-safe-separation-whitepaper/>
- [4] L. Reichstein, S. Schopferer and F. Jünger, "A Comparison of Command and Control Communication Protocols for Unmanned Aircraft: STANAG 4586 vs. MAVLink," 2022 International Conference on Unmanned Aircraft Systems (ICUAS), 2022, pp. 1283-1292, doi: 10.1109/ICUAS54217.2022.9836194.
- [5] Matzke, Barot, Ammann, Kessler, et al., "Addressing GNSS Vulnerabilities in AAM: A Multi-Modal UAV Testbed for Redundant and Reliable Navigation.", Uncrewed Aerial Vehicles in Geomatics (UAV-g), 2025.
- [6] DroC2om, Drone Critical Communication EU project, <https://cordis.europa.eu/article/id/415496-hybrid-satellite-and-5g-network-architecture-shaping-the-future-of-drone-communication>
- [7] ICAO, Performance-based communication and surveillance (pbc) manual, 2017.